with unrelenting ferocity. So powerful was the repulsive push of gravity that not only was the bang identified, it was revealed to be bigger—much bigger—than anyone had previously imagined. In the inflationary framework, the early universe expanded by an astonishingly huge factor compared with what is predicted by the standard big bang theory, enlarging our cosmological vista to a degree that dwarfed last century's realization that ours is but one galaxy among hundreds of billions.[2]

In this and the next chapter, we discuss inflationary cosmology. We will see that it provides a "front end" for the standard big bang model, offering critical modifications to the standard theory's claims about events during the universe's earliest moments. In doing so, inflationary cosmology resolves key issues that are beyond the reach of the standard big bang, makes a number of predictions that have been and in the near future will continue to be experimentally tested, and, perhaps most strikingly, shows how quantum processes can, through cosmological expansion, iron tiny wrinkles into the fabric of space that leave a visible imprint on the night sky. And beyond these achievements, inflationary cosmology gives significant insight into how the early universe may have acquired its exceedingly low entropy, taking us closer than ever to an explanation of the arrow of time.

Einstein and Repulsive Gravity

After putting the finishing touches on general relativity in 1915, Einstein applied his new equations for gravity to a variety of problems. One was the long-standing puzzle that Newton's equations couldn't account for the so-called precession of the perihelion of Mercury's orbit—the observed fact that Mercury does not trace the same path each time it orbits the sun: instead, each successive orbit shifts slightly relative to the previous. When Einstein redid the standard orbital calculations with his new equations, he derived the observed perihelion precession precisely, a result he found so thrilling that it gave him heart palpitations.[3] Einstein also applied general relativity to the question of how sharply the path of light emitted by a distant star would be bent by spacetime's curvature as it passed by the sun on its way to earth. In 1919, two teams of astronomers—one camped out on the island of Principe off the west coast of Africa, the other in Brazil— tested this prediction during a solar